MEDICAL INSURANCE HIGH-RISK PREDICTION

Machine Learning Mini Project | Random Forest Classification

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1. Project Overview

This project applies a Random Forest Classifier to a medical insurance dataset to predict whether an individual is marked as high-risk. The task is a supervised binary-classification problem: the model learns patterns from the input features and predicts the target category.

2. Dataset Description

The supplied CSV file contains **100,000 records and 54 columns**. It includes demographic, financial, health, insurance-plan and claims-related information. The target column is **is_high_risk**, where 0 represents not high-risk and 1 represents high-risk. The identifier **person_id** is not used as a predictive feature.

3. Selected Input Features

Feature	Description
age	Age of the individual
bmi	Body Mass Index
income	Individual income
annual_medical_cost	Annual medical cost
smoker	Smoking category
plan_type	Medical insurance plan category

4. Methodology

1. Load the CSV file using Pandas and inspect the data using head(), info() and describe().

2. Check missing values, duplicate records and numerical outliers; visualize selected variables with boxplots and a correlation heatmap.

3. Handle missing categorical values and convert categorical features into numerical form using one-hot encoding.

4. Separate the input features (X) from the target (y), then split the data into training and testing sets (80:20).

5. Train a RandomForestClassifier and generate predictions on the test data.

5. Model Evaluation

Evaluate the model using training and testing accuracy, a confusion matrix, and a classification report containing precision, recall and F1-score. These measures provide a more complete view than accuracy alone, especially if the two target classes are imbalanced. Hyperparameter tuning is optional and is not required for the basic implementation.

6. Expected Outcome and Limitation

The model produces a predicted high-risk or not-high-risk label for each test record. No accuracy or performance result is stated here because it must be calculated by running the notebook. This dataset-based exercise is for academic machine-learning practice only; its predictions should not be treated as medical advice or real insurance decisions.